Chapter 10. Models for Matched Pairs

McNemar, symmetry, agreement, Bradley-Terry

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Matched pairs (the same subject measured twice, a case matched with a control, two raters judging one slide) give dependent responses whose cross-classification is a square $I \times I$ table. We study marginal models, subject-specific (conditional) models, symmetry-type loglinear models and agreement.

10.1 Comparing dependent proportions

Motivation. Two proportions measured on the same subjects are correlated, so the independent-sample variance is wrong; the correct test of equal margins uses only the discordant pairs.

For n pairs let π_{ab} be the probability of outcome a for observation 1 and b for observation 2, n_{ab} the count, $p_{ab} = n_{ab}/n$, and $\{n_{ab}\} \sim \text{multinomial}(n; \{\pi_{ab}\})$.

Definition (Marginal homogeneity)

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all a.

For 2×2 tables $\pi_{1+} - \pi_{+1} = \pi_{12} - \pi_{21}$, so marginal homogeneity means $\pi_{12} = \pi_{21}$.

10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

Theorem 10.1 (Variance of a difference of dependent proportions)

Under multinomial sampling,

$$\begin{aligned} n\,\mathrm{Var}(d) &= \pi_{1+}(1-\pi_{1+}) + \pi_{+1}(1-\pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12}+\pi_{21}) - (\pi_{12}-\pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat\sigma(d)$ with $\hat\sigma^2(d) = \big[(p_{12}+p_{21}) - (p_{12}-p_{21})^2\big]/n$ is a large-sample confidence interval for δ.

Proof.

Step 1. Multinomial moments: $n\,\mathrm{Var}(p_{ab}) = \pi_{ab}(1-\pi_{ab})$, $n\,\mathrm{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells. Each margin is binomial, and

$$n\,\mathrm{Cov}(p_{11}+p_{21},\, p_{11}+p_{12}) = \pi_{11}(1-\pi_{11}-\pi_{12}-\pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21},$$

so $\mathrm{Var}(p_{+1}) + \mathrm{Var}(p_{1+}) - 2\,\mathrm{Cov}(p_{+1}, p_{1+})$ gives the first line.

Step 2. Directly, $n\,\mathrm{Var}(p_{21}-p_{12}) = \pi_{21}(1-\pi_{21}) + \pi_{12}(1-\pi_{12}) + 2\pi_{12}\pi_{21}$, which is the second line.

Step 3. By the multivariate CLT and consistency of $\hat\sigma(d)$, Slutsky's theorem gives the interval. ■

Under $H_0 : \delta = 0$ the variance becomes $(\pi_{12}+\pi_{21})/n$, estimated by $\hat\sigma_0^2(d) = (n_{12}+n_{21})/n^2$.

Theorem 10.2 (McNemar's test is the score test)

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is z_0^2, where

$$z_0 = \frac{d}{\hat\sigma_0(d)} = \frac{n_{21} - n_{12}}{(n_{21}+n_{12})^{1/2}}; \tag{10.1}$$

z_0^2 is asymptotically χ^2_1 under H_0 (**McNemar's test**).

Proof.

Step 1. Reparametrize by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12} + \pi_{21}$, $q = \pi_{21}/\sigma$, subject only to $\pi_{11} + \pi_{22} + \sigma = 1$; H_0 is $q = \frac{1}{2}$. With $n^* = n_{12} + n_{21}$,

$$L = n_{11} \log \pi_{11} + n_{22} \log \pi_{22} + n^* \log \sigma + n_{21} \log q + n_{12} \log(1 - q),$$

and q separates from the constrained parameters, so the information is block diagonal and the score test uses only the q-block.

Step 2. $U(q) = n_{21}/q - n_{12}/(1-q)$ and $\mathcal{I}(q) = n\sigma/[q(1-q)]$. At $q = \frac{1}{2}$, $\hat{\sigma} = n^*/n$: $U = 2(n_{21} - n_{12})$, $\mathcal{I} = 4n^*$, and $U^2/\mathcal{I} = (n_{21} - n_{12})^2/n^* = z_0^2$. Also $d/\hat{\sigma}_0(d) = [(n_{21} - n_{12})/n]/[(n^*)^{1/2}/n]$ gives (10.1); the χ^2_1 limit is standard score asymptotics. ■

The diagonal counts do not affect the test, though all pairs enter d and its standard error.

10.1.2 Prime Minister approval rating example

Example 10.3 (Prime Minister approval)

1600 British voters rated the Prime Minister twice, six months apart: $(n_{11}, n_{12}, n_{21}, n_{22}) = (794, 150, 86, 570)$ (1 = approve). Then $p_{1+} = 0.59$, $p_{+1} = 0.55$, $\hat{\sigma}(d) = 0.0095$, and a 95% interval for $\pi_{+1} - \pi_{1+}$ is $(-0.06, -0.02)$. McNemar's $z_0 = (86 - 150)/236^{1/2} = -4.17$: approval dropped.

10.1.3 Increased precision with dependent samples

Independent samples have $n \operatorname{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1 - \pi_1) + \pi_2(1 - \pi_2)$, the first form of Theorem 10.1 without the covariance term. Matched data usually have positive dependence ($\pi_{11}\pi_{22} > \pi_{12}\pi_{21}$), which reduces $\operatorname{Var}(d)$: in Example 10.3 independent samples would give SE 0.0175 instead of 0.0095.

10.1.4 Small-sample test comparing matched proportions

Proposition 10.4 (Exact conditional test)

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \text{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\text{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's z_0.

Proof.

Step 1. A multinomial conditioned on the total of a subset of cells is multinomial on that subset with renormalized probabilities; for the two cells $(1, 2), (2, 1)$ this gives the binomial, with parameter $\frac{1}{2}$ under H_0.

Step 2. With mean $n^*/2$ and variance $n^*/4$, $z = (n_{21} - \frac{1}{2}n^*)/(\frac{1}{4}n^*)^{1/2} = (n_{21} - n_{12})/(n^*)^{1/2} = z_0$. ■

For Example 10.3, $P(\text{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$ (one-sided).

10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Display each pair as its own 2×2 stratum (rows: occasion; columns: outcome; row totals $(1, 1)$). This $2 \times 2 \times n$ **subject-specific table** contrasts with the **population-averaged table** of pairs.

Equal success probability at both occasions for every subject means conditional independence given subject, which implies marginal homogeneity.

Theorem 10.5 (McNemar as a CMH statistic)

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2 \times 2 \times n$ subject-specific table equals $(n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof.

Step 1. Let $c_k \in \{0, 1, 2\}$ be the number of successes in stratum k. The hypergeometric moments are $E(n_{11k}) = c_k/2$ and $\mathrm{Var}(n_{11k}) = c_k(2 - c_k)/4$.

Step 2. Concordant strata ($c_k = 0, 2$) have zero variance and $n_{11k} = E(n_{11k})$, so they contribute nothing.

Step 3. Discordant strata have variance $\frac{1}{4}$ and $n_{11k} - \frac{1}{2} = +\frac{1}{2}$ (n_{12} strata) or $-\frac{1}{2}$ (n_{21} strata), so

$$\mathrm{CMH} = \frac{\left[\frac{1}{2}(n_{12} - n_{21})\right]^2}{\frac{1}{4}(n_{12} + n_{21})} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}}.$$

■

With more categories or occasions, generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4).

10.2 Conditional logistic regression for binary matched pairs

Motivation. A logit model with one parameter per subject, with those parameters conditioned away, yields the estimator n_{21}/n_{12} of a subject-specific odds ratio.

10.2.1 Marginal versus conditional models for matched pairs

Let (Y_1, Y_2) be a subject's pair (1 = success) and $x_1 = 0$, $x_2 = 1$.

Definition (Marginal and conditional models)

A **marginal** (population-averaged) model describes the margins, e.g. $\mathrm{logit}\, P(Y_t = 1) = \alpha + \beta x_t$. A **conditional** (subject-specific) model has link $P(Y_{it} = 1) = \alpha_i + \beta x_t$, $i = 1, \ldots, n$.

In the marginal logit model $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$; in the conditional model β is the association in the subject-stratified table.

Proposition 10.6 (Subject-specific versus population-averaged effects)

Let α_i be drawn from a distribution F in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1 + e^{a+bx_t})$ with $b = \beta$.

Proof.

(a) Average $(\alpha_i + \beta) - \alpha_i = \beta$ over F.

(b) The first statement is total probability. Take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. Then $E[\operatorname{expit}(\alpha)] = \frac{1}{2}$, so $b = \operatorname{logit}\left(\frac{1}{2}[\operatorname{expit}(c+\beta) + \operatorname{expit}(\beta - c)]\right) \to \operatorname{logit}(\frac{1}{2}) = 0$ as $c \to \infty$, while β is fixed; so $b \neq \beta$ in general. ■

10.2.2 A logit model with subject-specific probabilities

The Cox–Rasch model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad x_1 = 0,\ x_2 = 1, \tag{10.2}$$

so each subject's odds of success at occasion 2 are e^{β} times those at occasion 1, with responses independent given the parameters.

Proposition 10.7 (Heterogeneity induces positive association)

Suppose $\alpha_1, \ldots, \alpha_n$ are i.i.d. from F and, given α_i, Y_{i1}, Y_{i2} are independent with the probabilities of (10.2). Then $\operatorname{Cov}(Y_{i1}, Y_{i2}) \geq 0$ for every β, with equality if and only if F is degenerate.

Proof. [Supplement]

Step 1. With $f(a) = \operatorname{expit}(a)$, $g(a) = \operatorname{expit}(a + \beta)$, conditional independence gives $\operatorname{Cov}(Y_{i1}, Y_{i2}) = \operatorname{Cov}(f(\alpha), g(\alpha))$, $\alpha \sim F$.

Step 2. For an independent copy α', f, g strictly increasing give $(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha')) \geq 0$, whose expectation is $2\operatorname{Cov}(f(\alpha), g(\alpha))$.

Step 3. Equality forces $\alpha = \alpha'$ a.s., possible for independent copies only if F is degenerate. ■

So the model accounts for within-pair dependence, but its many α_i ruin ordinary ML (Proposition 10.9); the remedy is *conditional ML*.

10.2.3 Conditional ML inference for binary matched pairs

Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)

In model (10.2) let $S_i = y_{i1} + y_{i2}$.
(a) S_i is sufficient for α_i. Given $S_i = 0$ or $S_i = 2$ the pair is determined; given $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{\beta}}{1 + e^{\beta}}, \qquad P(Y_{i1} = 1, Y_{i2} = 0 \mid S_i = 1) = \frac{1}{1 + e^{\beta}}.$$

(b) The conditional likelihood is $\ell_c(\beta) = (e^{\beta})^{n_{21}}/(1 + e^{\beta})^{n^*}$ with $n^* = n_{12} + n_{21}$, i.e. $n_{21} \mid n^* \sim \operatorname{bin}(n^*, e^{\beta}/(1 + e^{\beta}))$.
(c) The conditional ML estimator and its standard error are

$$\hat{\beta} = \log \frac{n_{21}}{n_{12}}, \qquad \mathrm{SE} = \left(\frac{1}{n_{21}} + \frac{1}{n_{12}}\right)^{1/2}.$$

Proof.

Step 1. The joint mass function is proportional to $\exp\left(\sum_i \alpha_i S_i + \beta \sum_i y_{i2}\right)$, an exponential family with S_i sufficient for α_i.

Step 2. Given $S_i \in \{0, 2\}$ the pair is fixed. Given $S_i = 1$, with $c_i = (1 + e^{\alpha_i})(1 + e^{\alpha_i + \beta})$, $P(0,1) = e^{\alpha_i + \beta}/c_i$ and $P(1,0) = e^{\alpha_i}/c_i$, so $P(0, 1 \mid S_i = 1) = e^{\beta}/(1 + e^{\beta})$.

Step 3. Multiplying over the n^* discordant pairs gives $\ell_c(\beta) = e^{\beta n_{21}}/(1+e^{\beta})^{n^*}$, a binomial likelihood.

Step 4. $\log \ell_c$ is strictly concave with derivative $n_{21} - n^* e^{\beta}/(1+e^{\beta})$, zero at $e^{\hat{\beta}} = n_{21}/n_{12}$.

Step 5. The information $n^* e^{\beta}/(1+e^{\beta})^2$ equals $n_{21}n_{12}/n^*$ at $\hat{\beta}$; its inverse is $1/n_{21} + 1/n_{12}$. ■

The conditional test of $\beta = 0$ is the binomial test of Proposition 10.4.

Proposition 10.9 (Inconsistency of unconditional ML)

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of β in (10.2) is $2\log(n_{21}/n_{12})$, which converges to 2β.

Proof. [Supplement]

Step 1. A concordant pair is fitted perfectly by $\alpha_i \to \pm\infty$ (likelihood 1 for any β) and drops out.

Step 2. A $(1,0)$ pair has likelihood $p_1(1-p_2)$, $p_1 = \text{expit}(\alpha_i)$, $p_2 = \text{expit}(\alpha_i + \beta)$; the α_i-score $(1-p_1) - p_2$ vanishes at $\alpha_i = -\beta/2$, giving $\text{expit}(-\beta/2)^2$. A $(0,1)$ pair profiles to $\text{expit}(\beta/2)^2$.

Step 3. The profile log-likelihood $2n_{21}\log\text{expit}(\beta/2) + 2n_{12}\log\text{expit}(-\beta/2)$ is that of Theorem 10.8 in $\beta/2$, so $\hat{\beta}/2 = \log(n_{21}/n_{12})$, which tends to β by the law of large numbers. ■

10.2.4 Random effects in binary matched-pairs model

Treating $\alpha_i \sim N(\mu, \sigma^2)$ and integrating out gives a three-parameter GLMM (Chapter 12), which also yields $\hat{\beta} = \log(n_{21}/n_{12})$ when the sample log odds ratio is nonnegative.

Proposition 10.10 (Mantel–Haenszel estimator)

Applied to the n subject-specific 2×2 tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals n_{21}/n_{12}.

Proof. [Supplement] With rows (observation 2, observation 1) and columns (success, failure), $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, $n_{++k} = 2$, so

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1-y_{k1})}{\sum_k (1-y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

■

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	n_{11}	n_{12}
$Y_1 = 0$	n_{21}	n_{22}

grey: $S_i \in \{0, 2\}$, no information on β
orange: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^{\beta}/(1+e^{\beta})\big)$

Figure 10.1. Only the discordant cells carry conditional information about β.

10.2.5 Logistic regression for matched case–control studies

In a matched case–control study each case $(Y = 1)$ is matched with a control $(Y = 0)$ and both are measured on X. With

$$\text{logit}\, P(Y_{it} = 1) = \alpha_i + \beta x_{it}, \tag{10.3}$$

e^{β} is an XY odds ratio and hence estimable from retrospective data.

Example 10.11 (Diabetes and myocardial infarction)

144 Navajo MI cases were matched with 144 controls on age and gender. In 37 pairs only the case had diabetes, in 16 only the control, in 9 both and in 82 neither; so $e^{\hat\beta} = 37/16 = 2.3$, $\hat\beta = 0.84$.

10.2.6 Conditional ML for matched pairs with multiple predictors

With p predictors the model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1 + e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is obtained by fitting an ordinary logistic regression, with no intercept, to the discordant pairs only, with response $y_i^* = 1$ for $(0,1)$ pairs, $y_i^* = 0$ for $(1,0)$ pairs and predictor x_i^*;
(b) a predictor with $x_{ji1} = x_{ji2}$ for all i (a between-pair effect) cancels from the conditional likelihood, so its coefficient cannot be estimated by conditional ML.

Proof.

Step 1. As in Theorem 10.8, the factors e^{α_i} and $c_i = (1 + e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1 + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$ cancel, leaving $e^{x_{i2}'\boldsymbol{\beta}}/(e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}})$; divide by $e^{x_{i1}'\boldsymbol{\beta}}$.

Step 2. The conditional likelihood $\prod_{S_i=1} \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*}\left[1 - \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})\right]^{1-y_i^*}$ is the no-intercept logistic likelihood, proving (a).

Step 3. If $x_{ji1} = x_{ji2}$ for all i, every x_i^* has jth component 0 and β_j disappears, proving (b). ■

In Example 10.11 (control $t = 1$, case $t = 2$) $y_i^* = 1$ and $x_i^* = -1$ (16 pairs) or $+1$ (37 pairs), again giving $\hat\beta = \log(37/16)$.

10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and to matched sets or clusters. By Theorem 10.12(b), conditional ML estimates only within-cluster effects; random-effects models (Chapter 12) have no such restriction.

10.3 Marginal models for square contingency tables

Motivation. For $I > 2$ categories the two margins are compared by models acting directly on them: cumulative logits for ordinal and baseline-category logits for nominal responses.

For a square table with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$ for all a.

10.3.1 Marginal models for ordinal classifications

The cumulative logit model

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

($x_1 = 0$, $x_2 = 1$) has proportional odds; $\beta > 0$ means Y_1 tends to be higher, and marginal homogeneity is $\beta = 0$. ML fitting maximizes the multinomial likelihood in $\{\pi_{ab}\}$ subject to constraints on the margins; the fit has df $= 2(I-1) - I = I - 2$.

10.3.2 Premarital and extramarital sex example

Example 10.13 (Premarital and extramarital sex)

In the 1989 GSS, 475 subjects rated premarital and extramarital sex on 1 = always wrong, ..., 4 = not wrong at all. The cumulative marginal proportions are $(0.307, 0.389, 0.611)$ (premarital) and $(0.815, 0.918, 0.987)$ (extramarital). Model (10.5) gives $\hat\beta = 2.51$ (SE 0.13): far more permissive views on premarital sex. Marginal homogeneity has $G^2 = 348.1$ (df = 3), model (10.5) $G^2 = 35.1$ (df = 2); see Section 10.4.7 for better models.

10.3.3 Marginal models for nominal classifications

The baseline-category model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

is saturated for the margins; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$, tested by G^2 or X^2 with df $= I - 1$. A non-iterative alternative uses $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$.

Proposition 10.14 (Bhapkar and Stuart statistics)

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \ne b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically χ^2_{I-1}. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat v_{ab0} = -(p_{ab} + p_{ba})$, $\hat v_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, W_0 is McNemar's statistic z_0^2.

Proof. [Supplement]

Step 1. $\mathbf{d}$ is the mean of i.i.d. $\mathbf{X}_k$ with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \ne b$, $E(X_aX_b) = -(\pi_{ab} + \pi_{ba})$, and $E(X_a^2) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$; subtracting $E(X_a)E(X_b)$ gives (a).

Step 2. By the CLT and consistency of $\hat{\mathbf{V}}$, under H_0 ($E\mathbf{d} = \mathbf{0}$) $W \xrightarrow{d} \chi^2_{I-1}$.

Step 3. $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By Sherman–Morrison with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q},$$

so $W = W_0/(1 - W_0/n)$.

Step 4. For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{12} + p_{21}$, so $W_0 = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ■

For ordered categories a single-df ordinal test is often more powerful.

10.3.4 Migration example

Example 10.15 (Migration 1980–1985)

Region of residence (NE, MW, S, W) in 1980 and 1985 for 55,981 U.S. residents; about 95% lie on the diagonal, and, e.g., 124 moved NE→W but 63 W→NE. Marginal homogeneity has $G^2 = 240.8$ (df = 3), Bhapkar's $W = 236.5$. The marginal proportions $(0.218, 0.260, 0.330, 0.191)$ in 1980 and $(0.213, 0.253, 0.339, 0.194)$ in 1985 differ by -0.005, -0.007, 0.009, 0.003 (each ± 0.001 by Theorem 10.1): significant only because n is huge.

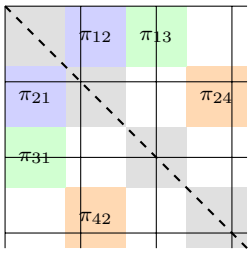
10.4 Symmetry, quasi-symmetry, and quasi-independence

Motivation. Loglinear models for the joint table: quasi-symmetry fits well, has a subject-specific meaning and yields a likelihood-ratio test of marginal homogeneity.

Definition (Symmetry)

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity ($\pi_{a+} = \sum_b \pi_{ba} = \pi_{+a}$); for $I > 2$ the converse fails.



grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a, b), (b, a)$: equal under S; under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

Figure 10.2. Mirror pairs of a square table ($I = 4$).

10.4.1 Symmetry as logit and loglinear models

For $\pi_{ab} > 0$, symmetry is $\log(\pi_{ab}/\pi_{ba}) = 0$ $(a < b)$, or, with $\mu_{ab} = n\pi_{ab}$,

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.7}$$

Theorem 10.16 (Fitting and testing symmetry)

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's z_0^2;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

Proof.

Step 1. (a) Writing the model as $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood splits into $(n_{ab} + n_{ba}) \log \mu - 2\mu$ for each pair $a < b$ and $n_{aa} \log \mu_{aa} - \mu_{aa}$ for each diagonal cell; maximize each. The multinomial fit is the same.

Step 2. (b) df $= I^2 - I - \binom{I}{2} = I(I-1)/2$.

Step 3. (c) With $m = (n_{ab}+n_{ba})/2$, both cells of a pair have $(n-m)^2 = (n_{ab}-n_{ba})^2/4$, contributing $2(n_{ab} - n_{ba})^2/(4m)$; diagonal cells contribute 0.

Step 4. (d) Each pair is fitted by one parameter with design $(1, 1)'$ and equal weights, so the leverage is $h = \frac{1}{2}$ and $r_{ab} = (n_{ab} - m)/\sqrt{m/2} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2}$; squaring and summing gives (c). ■

10.4.2 Quasi-symmetry

Definition (Quasi-symmetry)

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is $\lambda_a^X = \lambda_a^Y$; independence is $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$, QS can be written

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

Proposition 10.17 (Likelihood equations for quasi-symmetry)

The ML fit of QS satisfies

$$\hat{\mu}_{a+} = n_{a+}, \qquad \hat{\mu}_{+b} = n_{+b}, \qquad \hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat{\mu}_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

Proof.

Step 1. Poisson likelihood equations equate the sufficient statistics (row totals, column totals, symmetric sums $n_{ab} + n_{ba}$, $a \le b$) to their expectations; $a = b$ gives $\hat{\mu}_{aa} = n_{aa}$.

Step 2. Given the third set and the row equations, $\hat{\mu}_{+a} = \sum_b (n_{ab} + n_{ba} - \hat{\mu}_{ab}) = n_{a+} + n_{+a} - \hat{\mu}_{a+} = n_{+a}$.

Step 3. Form (10.9) has $I(I+1)/2 + (I-1)$ parameters, so df $= (I-1)(I-2)/2$. ■

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with symmetry the case $\alpha_a = \beta_a$.

Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)

Let all $\pi_{ab} > 0$.
(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

Proof. [Supplement]

Step 1. ($\Rightarrow$) Under (10.10) the α's and β's cancel, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$, equal by symmetry of γ.

Step 2. ($\Leftarrow$) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$, $\gamma_{ab} = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$; then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ and (10.11) says $\gamma_{ab} = \gamma_{ba}$.

Step 3. (b) Similarly $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$, which is unchanged when a, b are swapped since γ is symmetric. ■

The meaning of QS comes from the subject-specific version of (10.6):

$$\log\frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)

If (10.12) holds and subjects are sampled from any distribution F of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

Proof. [Supplement]

Step 1. With $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$, $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k+\beta_k}$, conditional independence gives $P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = e^{\alpha_a}e^{\alpha_b+\beta_b}/[D_1(\boldsymbol{\alpha})D_2(\boldsymbol{\alpha})]$.

Step 2. Averaging over F, $\pi_{ab} = e^{\beta_b}\gamma_{ab}$ with $\gamma_{ab} = E_F[e^{\alpha_a+\alpha_b}/(D_1D_2)] = \gamma_{ba}$. This is (10.10) with $\alpha_a \equiv 1$, i.e. (10.9) with $\tau_b = \beta_b$. ■

Moreover (Conaway 1989) the conditional ML estimates of $\{\beta_j\}$ equal $\hat\lambda_j^Y - \hat\lambda_j^X$ from the QS fit.

10.4.3 Quasi-independence

Definition (Quasi-independence)

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log\mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a\, I(a = b). \tag{10.13}$$

The likelihood equations $\hat\mu_{a+} = n_{a+}$, $\hat\mu_{+a} = n_{+a}$, $\hat\mu_{aa} = n_{aa}$ fit the diagonal exactly; off-diagonal rectangle odds ratios equal 1, and df $= (I-1)^2 - I$ $(I \geq 3)$.

Proposition 10.20 (Quasi-independence within quasi-symmetry)

QI is the special case of QS in which the γ_{ab}, $a \neq b$, in (10.10) are all equal. For $I = 3$ the two models are equivalent.

Proof. [Supplement]

Step 1. QI is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$; conversely a constant off-diagonal γ is absorbed into α, and γ_{aa} into δ_a.

Step 2. For $I = 3$, the positive numbers $c_1 = (\gamma_{12}\gamma_{13}/\gamma_{23})^{1/2}$, $c_2 = (\gamma_{12}\gamma_{23}/\gamma_{13})^{1/2}$, $c_3 = (\gamma_{13}\gamma_{23}/\gamma_{12})^{1/2}$ satisfy $c_a c_b = \gamma_{ab}$ $(a \neq b)$. Then $\pi_{ab} = (\alpha_a c_a)(\beta_b c_b)$ off the diagonal, which is QI; with Step 1 the models coincide. ■

10.4.4 Migration revisited

Example 10.21 (Migration: loglinear models)

For Example 10.15, independence ($G^2 = 125{,}923$) and symmetry ($G^2 = 243.6$, df $= 6$) fail; QI is better ($G^2 = 69.5$, df $= 5$) but misfits NE→S and W→S; QS fits well ($G^2 = 3.0$, df $= 3$). Its effects are $\hat{\tau} = (-0.672, -0.623, 0.122)$ (W baseline): a person's odds of living in the South rather than the West were $e^{0.122} = 1.13$ times higher in 1985 than in 1980.

10.4.5 Marginal homogeneity and quasi-symmetry

Theorem 10.22 (Caussinus)

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

Proof.

Step 1. ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$ and implies marginal homogeneity.

Step 2. ($\Rightarrow$) Write $\pi_{ab} = \kappa_a \gamma^*_{ab}$ with $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a\beta_b\gamma_{ab} = \gamma^*_{ba} > 0$. Marginal homogeneity $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$ reads

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1.$$

Step 3. For j with $\kappa_j = \min_a \kappa_a$, a positive-weight average equal to the minimum forces all $\kappa_a = \kappa$. Then $\pi_{ab} = \kappa\gamma^*_{ab} = \pi_{ba}$. ■

Corollary 10.23 (Conditional test of marginal homogeneity)

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda^X_a = \lambda^Y_a$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

Proof. The equivalence is Theorem 10.22. In (10.9) symmetry is the nested submodel $\tau_1 = \cdots = \tau_{I-1} = 0$, so the deviance difference is the LR statistic, with df from Theorem 10.16(b) and Proposition 10.17. ■

For the migration data $G^2(S \mid QS) = 240.6$ (df $= 3$), close to $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

10.4.6 Ordinal quasi-symmetry model

With scores $u_1 \le \cdots \le u_I$, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

QS with $\tau_b = \beta u_b$; symmetry is $\beta = 0$. Equivalently

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b, \tag{10.15}$$

a no-intercept logit model for $n_{ab} \mid (n_{ab} + n_{ba})$ with predictor $u_b - u_a$.

Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat\mu_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat\mu_{+b} = \sum_b u_b n_{+b}$ and $\hat\mu_{ab} + \hat\mu_{ba} = n_{ab} + n_{ba}$ $(a \le b)$. The fitted margins need not equal the observed ones, but the fitted row and column means equal the observed means.
(b) Under (10.14), the column mean exceeds the row mean if $\beta > 0$ and is smaller if $\beta < 0$; hence ordinal QS + marginal homogeneity = symmetry, and $H_0 : \beta = 0$ tests marginal homogeneity (df $= 1$).

Proof. [Supplement]

Step 1. (a) The sufficient statistics of $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ are the symmetric sums and $\sum_b u_b n_{+b}$, giving the second and third sets. Summing $(u_a + u_b)$ times the third set over $a \le b$ gives $\sum_{a,b}(u_a + u_b)\hat\mu_{ab} = \sum_{a,b}(u_a + u_b) n_{ab}$; subtracting the second set gives the first.

Step 2. (b) By (10.15),

$$\sum_{a,b}(u_b - u_a)\pi_{ab} = \sum_{a<b}(u_b - u_a)\pi_{ba}\left(e^{\beta(u_b - u_a)} - 1\right),$$

every term having the sign of β. Equal means force $\beta = 0$, i.e. symmetry; the converse is immediate. ■

10.4.7 Premarital and extramarital sex revisited

Example 10.25 (Premarital and extramarital sex: symmetry models)

For Example 10.13, symmetry fails ($G^2 = 402.2$, df $= 6$); QS ($G^2 = 1.4$, df $= 3$) and ordinal QS with scores $1, \ldots, 4$ ($G^2 = 2.1$, df $= 5$, $\hat\beta = -2.86$) fit well. By (10.15), a premarital response x categories more permissive than the extramarital one is $e^{2.86x}$ times as likely as the reverse.

10.4.8 Other ordinal models for square tables

The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

describes $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse); symmetry is $\tau = 0$. The **quasi-uniform association** model

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b) \tag{10.17}$$

adds linear-by-linear association off the diagonal to QI; it is a special case of QS. For Example 10.13, $\hat{\tau} = -4.130$ (SE 0.451) and $\hat{\beta} = 0.632$ (SE 0.106).

10.5 Measuring agreement between observers

Motivation. Agreement between two raters is a different property from association and needs diagonal-oriented models and indices such as kappa.

Example 10.26 (Pathologists' diagnoses)

Two pathologists A and B classified 118 slides into four ordered categories (1 = negative, ..., 4 = invasive carcinoma). The diagonal counts are $(22, 7, 36, 10)$; the most common disagreements are B rating 3 when A rated 2 (14) or 4 (17). Independence fits poorly ($G^2 = 118.0$, df $= 9$), with large positive diagonal residuals.

10.5.1 Agreement: departures from independence

With π_{ab} the probability that A chooses a and B b, $\sum_a \pi_{aa}$ is the agreement probability. Strong agreement requires strong association but not conversely (A might always rate one category higher). Independence is the baseline of chance agreement.

10.5.2 Using quasi-independence to analyze agreement

For two subjects rated a and b by one observer and b, a or a, b by the other, the odds of agreement are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

Proposition 10.27 (Agreement odds under QI and QS)

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

Proof. [Supplement] In $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ the main effects $\lambda_a^X + \lambda_a^Y + \lambda_b^X + \lambda_b^Y$ cancel against those of the off-diagonal cells, leaving $\delta_a + \delta_b$ under (10.13) and $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$ under (10.8). ■

For Example 10.26, QI has $G^2 = 13.2$ (df $= 5$) and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

10.5.3 Quasi-symmetry and agreement modeling

QS also permits association given disagreement and fits Example 10.26 well ($G^2 = 1.0$, df $= 2$, $\hat{\tau}_{23} = 10.7$). Pairs with $n_{ab} + n_{ba} = 0$ are fitted by 0 and should be removed for correct df. Symmetry fits poorly ($G^2 = 39.2$, df $= 5$), and $G^2(S \mid QS) = 38.2$ (df $= 3$) shows strong marginal heterogeneity.

10.5.4 Kappa measure of agreement

Definition (Cohen's kappa)

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

$\kappa = 0$ at chance agreement, $\kappa = 1$ at perfect agreement.

Theorem 10.28 (Asymptotic variance of sample kappa)

Under multinomial sampling, $\sqrt{n}(\hat\kappa - \kappa) \xrightarrow{d} N(0, \sigma^2_\kappa)$, where

$$\sigma^2_\kappa = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)\left[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})\right]}{(1-P_e)^3} + \frac{(1-P_o)^2\left[\sum_a\sum_b \pi_{ab}(\pi_{b+}+\pi_{+a})^2 - 4P_e^2\right]}{(1-P_e)^4}.$$

The estimated variance of $\hat\kappa$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions and divides by n.

Proof. [Supplement]

Step 1. By the delta method for the multinomial, $\sigma^2 = \sum_{a,b}\pi_{ab}g_{ab}^2 - \left(\sum_{a,b}\pi_{ab}g_{ab}\right)^2$ with $g_{ab} = \partial\kappa/\partial\pi_{ab}$.

Step 2. With $\kappa = 1 - (1-P_o)/(1-P_e)$, $\partial P_o/\partial\pi_{ab} = I(a=b)$ and $\partial P_e/\partial\pi_{ab} = \pi_{+a} + \pi_{b+}$,

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a}+\pi_{b+})}{(1-P_e)^2}.$$

Step 3. Since $\sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+}) = 2P_e$, $\sum \pi_{ab}g_{ab} = [P_o(1-P_e) - 2P_e(1-P_o)]/(1-P_e)^2$. With $s = \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})$ and $t = \sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+})^2$,

$$(1-P_e)^4\sum_{a,b}\pi_{ab}g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

Step 4. Subtracting the square of the first moment,

$$(1-P_e)^4\sigma^2_\kappa = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t - 4P_e^2),$$

which is the stated formula. ∎

For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat\kappa = 0.493$ (SE 0.057).

10.5.5 Weighted kappa: quantifying disagreement

Definition (Weighted kappa)

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a\sum_b w_{ab}\pi_{ab} - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1 - |a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1 - (a-b)^2/(I-1)^2$.

Both kappas depend strongly on the margins, so models of the agreement structure are more informative than a single index.

10.5.6 Extensions to multiple observers

With R observers one can model the $\binom{R}{2}$ pairwise marginal tables simultaneously, use generalized kappas, or use latent-class models (Chapter 13).